

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283770

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

YAMASAKI; Tatsuya et al.

PRESSURE SENSOR

Abstract

The present invention provides a pressure sensor in which a decrease in sensitivity or leakage of a fluid can be prevented even when the pressure sensor receives pressure from the fluid. This pressure sensor according to the present invention comprises a housing having a space inside that is a fluid flow channel, a sensor chip that comprises a strain detection part and is constituted from a thin-film semiconductor, and a strain transmission material that is provided inside the housing. The strain transmission material comprises a pressure-receiving surface in contact with the space that is a flow channel, and a bonding surface that is positioned on the opposite side of the strain transmission material from the pressure-receiving surface and is fixed to an inside surface of the housing. The entire sensor chip is embedded inside the strain transmission material.

Inventors: YAMASAKI; Tatsuya (Tokyo, JP), KANAMARU; Masatoshi (Tokyo, JP), AONO; Takanori (Tokyo, JP), SUZUKI; Yoichiro (Tokyo, JP)

Applicant: Hitachi High-Tech Corporation (Tokyo, JP)

Family ID: 1000008669191

Assignee: Hitachi High-Tech Corporation (Tokyo, JP)

Appl. No.: 18/859386

Filed (or PCT Filed): April 06, 2023

PCT No.: PCT/JP2023/014233

Foreign Application Priority Data

JP 2022-088556

May. 31, 2022

Publication Classification

Int. Cl.: G01L19/00 (20060101); **G01L9/00** (20060101); **G01L19/06** (20060101)

U.S. Cl.:

CPC G01L19/0023 (20130101); **G01L9/0051** (20130101); **G01L19/0654** (20130101);
G01L19/0672 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a pressure sensor, particularly to a pressure sensor that can be provided in a dispensing device.

BACKGROUND ART

[0002] A pressure sensor for measuring a pressure of a fluid is implemented by, for example, a thin-film semiconductor strain sensor, and is used in a dispensing device. The dispensing device is provided in, for example, an automatic analyzer, and aspirates or discharges a sample (specimen) or a reagent which is a liquid by a nozzle. The pressure sensor provided in the dispensing device may be used to detect an abnormality such as clogging or air suction of the nozzle.

[0003] PTL 1 describes an example of a pressure sensor in the related art that measures a pressure of a fluid. The pressure sensor disclosed in PTL 1 includes a sensor chip supported by a base plate. The sensor chip includes a sensor diaphragm that generates strain according to the pressure of the fluid, and detects the pressure. The base plate is bonded to a support diaphragm. A center portion of the support diaphragm includes a pressure introduction hole for introducing the pressure of the fluid to the sensor chip. A pressure-receiving portion of the sensor chip is positioned in the same direction as a bonding portion of the sensor chip to the base plate with respect to the sensor chip.

CITATION LIST

Patent Literature

[0004] PTL 1: JP2006-003234A

SUMMARY OF INVENTION

Technical Problem

[0005] In the pressure sensor in the related art that is implemented by the thin-film semiconductor strain sensor, as described in PTL 1, the sensor chip is generally fixed to a housing (or a base) of the pressure sensor, and the bonding portion of the sensor chip to the housing and the pressure-receiving portion of the sensor chip are generally at positions in the same direction with respect to the sensor chip. In this structure, a fluid flow channel is formed in the housing, and not only the pressure-receiving portion of the sensor chip but also the bonding portion of the sensor chip to the housing is in direct contact with the fluid. Therefore, when the bonding portion of the sensor chip to the housing receives the pressure of the fluid, a tensile stress is generated in a direction in which the bonding portion is peeled off from the housing, and the bonding portion may be peeled off from the housing. The peeling off of the bonding portion of the sensor chip to the housing causes a decrease in sensitivity of the pressure sensor and a leakage of the fluid from the bonding portion.

As described above, in the pressure sensor in the related art, there is a concern that when the pressure from the fluid is received, the tensile stress is generated in the bonding portion of the sensor chip to the housing, which causes the decrease in the sensitivity or the leakage of the fluid.

[0006] An object of the invention is to provide a pressure sensor in which a decrease in sensitivity or a leakage of a fluid can be prevented even when the pressure sensor receives a pressure from the fluid.

Solution to Problem

[0007] A pressure sensor according to the invention includes: a housing having therein a space serving as a fluid flow channel; a sensor chip including a strain detection unit and formed of a thin-film semiconductor; and a strain transmission material provided inside the housing. The strain transmission material includes a pressure-receiving surface in contact with the space, and a bonding surface positioned at an opposite side of the strain transmission material from the pressure-receiving surface and fixed to an inner surface of the housing. The entire sensor chip is embedded in the strain transmission material.

Advantageous Effects of Invention

[0008] According to the invention, it is possible to provide a pressure sensor in which a decrease in sensitivity or a leakage of a fluid can be prevented even when the pressure sensor receives a pressure from the fluid.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a diagram showing a basic configuration of a dispensing device including a pressure sensor according to Embodiment 1 of the invention.

[0010] FIG. 2 is a diagram showing an internal state of a pipe in an arm immediately after a nozzle aspirates a liquid.

[0011] FIG. 3 is a schematic view showing an example of a cross section of a pressure sensor in the related art.

[0012] FIG. 4A is an exploded perspective view of the pressure sensor according to Embodiment 1.

[0013] FIG. 4B is a cross-sectional view of the pressure sensor according to Embodiment 1 taken along a plane orthogonal to a direction of a flow channel.

[0014] FIG. 5 is an exploded perspective view showing only a sensor chip, a bonding wire, and a lead frame in the pressure sensor according to Embodiment 1.

[0015] FIG. 6 is an exploded perspective view showing only a sensor chip, a bonding wire, a lead frame, and a cap member in a pressure sensor according to Embodiment 2 of the invention.

DESCRIPTION OF EMBODIMENTS

[0016] A pressure sensor according to the invention includes a housing constituting a fluid flow channel and a sensor chip provided inside the housing, and even when the pressure sensor receives a pressure from a fluid, a tensile stress is not generated in a bonding portion of the sensor chip to the housing, so that a decrease in sensitivity or a leakage of the fluid can be prevented. Therefore, the invention can provide a highly reliable pressure sensor.

[0017] Hereinafter, a pressure sensor according to an embodiment of the invention will be described with reference to the drawings. In the drawings used in the present specification, the same or corresponding components are denoted by the same reference numerals, and repeated description of these components may be omitted.

Embodiment 1

[0018] A pressure sensor according to Embodiment 1 of the invention will be described with reference to FIGS. 1 to 5. Hereinafter, a configuration in which the pressure sensor according to the present embodiment is provided in a sample dispensing mechanism of a dispensing device will be described as an example.

[0019] FIG. 1 is a diagram showing a basic configuration of a dispensing device 1 including a pressure sensor 15 according to the present embodiment. The dispensing device 1 includes a flow channel system, a sample dispensing mechanism 13, and a control unit 14.

[0020] The flow channel system of the dispensing device 1 includes a nozzle 2, a syringe pump 4, a solenoid valve 5, a gear pump 6, and a tank 7 that stores system water, and these components are connected by a pipe 8. The sample dispensing mechanism 13 includes an arm 16 including a nozzle

2, and aspirates and discharges a liquid (for example, a sample or the system water) through the nozzle 2. The control unit 14 controls the dispensing device 1. For example, a motor for driving the sample dispensing mechanism 13 or the like is controlled by the control unit 14. The syringe pump 4 includes a container 9, a plunger 10, a ball screw 11, and a drive motor 12. The drive motor 12 is controlled by the control unit 14.

[0021] The arm 16 includes the pressure sensor 15 according to the present embodiment therein. Since the arm 16 moves the nozzle 2 to a position where the liquid is aspirated and discharged, the arm 16 can be rotationally moved and moved up and down.

[0022] FIG. 2 is a diagram showing an internal state of the pipe 8 in the arm 16 immediately after the nozzle 2 aspirates the liquid.

[0023] An inside of the pipe 8 is filled with system water 21 which is water for syringe pressure transmission. A pressure due to the syringe pump 4 is transmitted into the pipe 8, so that the nozzle 2 can aspirate or discharge a liquid 22 (for example, the sample or the system water).

[0024] When the nozzle 2 aspirates the liquid 22, the plunger 10 in the syringe pump 4 is pulled from the container 9 while the solenoid valve 5 is in a closed state. When the nozzle 2 discharges the liquid 22, the plunger 10 in the syringe pump 4 is pushed into the container 9 while the solenoid valve 5 is in the closed state. When the nozzle 2 aspirates the liquid 22 such as the sample, the nozzle 2 aspirates the liquid 22 after aspirating dividing air 23 for division, so that the liquid 22 does not mix with the system water 21 in the pipe 8.

[0025] After the nozzle 2 discharges the liquid 22, cleaning of the nozzle 2 is performed. During the cleaning of the nozzle 2, the system water 21 inside the nozzle 2 is pushed out of the nozzle 2 simultaneously with application of cleaning water to an outer wall of the nozzle 2. When the system water 21 is pushed out of the nozzle 2 during the cleaning of the nozzle 2, the solenoid valve 5 is opened to utilize a pressure of the gear pump 6. Therefore, during the cleaning of the nozzle 2, the system water 21 is pushed out under a pressure higher than that when being pushed out by the syringe pump 4.

[0026] In the dispensing device 1, the pressure sensor 15 is provided on the pipe 8 in order to detect an abnormality such as clogging or air suction of the nozzle 2, which may occur during a dispensing operation. The pressure sensor 15 monitors a pressure of the system water 21, and detects a pressure change occurring when the abnormality occurs in the nozzle 2.

[0027] An arrangement position of the pressure sensor 15 can be freely determined. In the present embodiment, as an example, in order to sense the pressure change of the nozzle 2 sensitively, the pressure sensor 15 is provided inside the arm 16 at a position relatively close to the nozzle 2. The pressure sensor 15 is not limited to being provided inside of the arm 16, and may be provided on, for example, a side surface of the sample dispensing mechanism 13.

[0028] A structure of a general pressure sensor in the related art and problems thereof will be described with reference to FIG. 3.

[0029] FIG. 3 is a schematic view showing an example of a cross section of a pressure sensor 15a in the related art. The pressure sensor 15a includes a sensor chip 31 formed of a thin-film semiconductor and a housing 32 to which the sensor chip 31 is bonded, and a fluid flowing through a flow channel 35 is a measurement target. A strain gauge portion 31b is provided on a surface of a center portion of the sensor chip 31. The cross section shown in FIG. 3 is a cross section taken along a plane passing through the strain gauge portion 31b and along a normal direction of the sensor chip 31. FIG. 3 shows the sensor chip 31 deformed by receiving a pressure p from the fluid.

[0030] The sensor chip 31 is disposed to receive the pressure p of the fluid flowing through the flow channel 35, and is bonded to the housing 32 by a bonding agent 40. The fluid flowing through the flow channel 35 as the measurement target is, for example, the system water 21. The sensor chip 31 is further deformed by the pressure p of the fluid as the measurement target, and measures the pressure p of the fluid as the measurement target by measuring strain due to the deformation. The strain is measured by the strain gauge portion 31b.

[0031] In the pressure sensor **15a**, when the sensor chip **31** is deformed by receiving the pressure **p** from the fluid as the measurement target, a force in a direction in which the joint is peeled off is generated on a bonding surface between the sensor chip **31** and the bonding agent **40** and a bonding surface between the housing **32** and the bonding agent **40**. For this reason, in the pressure sensor **15a** in the related art, there is a concern that the sensor chip **31** may be peeled off from the housing **32** when the sensor chip **31** receives an excessive pressure or a repeated pressure. Further, in the pressure sensor **15a** in the related art, since a part of the surface of the sensor chip **31** or the bonding surface described above is in contact with the fluid as the measurement target or outside air, there is a concern that deterioration of the bonding surface or a decrease in a bonding force may occur due to an influence of a humidity on the fluid as the measurement target or the outside air.

[0032] The pressure sensor **15** according to the present embodiment can solve these problems. A structure and characteristics of the pressure sensor **15** according to the present embodiment will be described with reference to FIGS. **4A**, **4B**, and **5**.

[0033] FIGS. **4A** and **4B** are diagrams showing an example of the structure of the pressure sensor **15** according to the present embodiment. FIG. **4A** is an exploded perspective view of the pressure sensor **15**. FIG. **4B** is a cross-sectional view of the pressure sensor **15** taken along a plane orthogonal to an extending direction of a flow channel.

[0034] The pressure sensor **15** includes the sensor chip **31** formed of the thin-film semiconductor, the housing **32** accommodating the sensor chip **31**, and a strain transmission material **36** provided inside the housing **32**, and a fluid flowing through a flow channel **33** is the measurement target. The fluid flow channel **33** is a space formed inside the housing **32**. The strain gauge portion **31b** as a strain detection unit is provided on the surface of the center portion of the sensor chip **31**. The fluid flowing through the flow channel **33** as the measurement target is, for example, the system water **21**.

[0035] The cross section shown in FIG. **4B** is a cross section taken along a plane (plane along the normal direction of the sensor chip **31**) passing through the strain gauge portion **31b** and orthogonal to a direction in which the fluid flows through the flow channel **33**.

[0036] Here, the normal direction of the sensor chip **31** is defined as a “upper-lower direction”, an extending direction of the flow channel **33** (the direction in which the fluid flows through the flow channel **33**) is defined as a “flow channel direction”, and a direction orthogonal to the upper-lower direction and the flow channel direction is defined as a “horizontal direction”.

[0037] The housing **32** includes an upper housing **32a** and a lower housing **32b** positioned below the upper housing **32a**, and includes, in the housing **32**, the flow channel **33** through which the system water **21** flows. The upper housing **32a** and the lower housing **32b** are in contact with each other, and a contact surface **32c** of the upper housing **32a** and the lower housing **32b** is sealed with a sealing member, so that the system water **21** does not leak from the housing **32**. For example, an O-ring or a sealing agent can be used as the sealing member. It should be noted that it is not always necessary for the housing **32** to be constituted by two parts (the upper housing **32a** and the lower housing **32b**), and for example, the housing **32** may be configured with one part, for example, by insert molding, or may be configured with three or more parts.

[0038] Screw portions (not shown) are provided at both ends of the flow channel **33** in the flow channel direction. The flow channel **33** can be connected to the pipe **8** of the dispensing device **1** via a joint by the screw portion.

[0039] The strain transmission material **36** is fixed to an inner surface of the housing **32** (that is, a wall surface of the flow channel **33**). A surface of the strain transmission material **36** fixed to the inner surface of the housing **32** is referred to as a bonding surface **36b**. When coming into contact with the system water **21** flowing through the flow channel **33**, the strain transmission material **36** is compressed by a pressure of the system water **21**. A surface of the strain transmission material **36** in contact with the system water **21** is referred to as a pressure-receiving surface **36a**. The pressure-receiving surface **36a** is a surface that is in contact with the space that is the fluid flow channel **33**

and receives the pressure of the system water **21**. The bonding surface **36b** is positioned at an opposite side of the strain transmission material **36** from the pressure-receiving surface **36a**. The strain transmission material **36** can be made of a flexible elastic body, and can be made of a gel substance such as silicone gel.

[0040] The pressure sensor **15** may include a lead frame **45** electrically connected to the sensor chip **31**. The lead frame **45** is provided to be in contact with the sensor chip **31** in the upper-lower direction. Note that a upper-lower relationship between positions of the lead frame **45** and the sensor chip **31** can be freely determined, and either the lead frame **45** or the sensor chip **31** may be provided on an upper side (position close to the inner surface of the housing **32** in FIG. **4B**). In the present embodiment, as an example, the sensor chip **31** is positioned on the upper side of the lead frame **45**.

[0041] The entire sensor chip **31** and a part of the lead frame **45** are embedded inside the strain transmission material **36**.

[0042] The lead frame **45** includes an aperture **45a** which is an opening (hole) penetrating the lead frame **45**. The aperture **45a** opens in a range facing a portion of the sensor chip **31** including the strain gauge portion **31b**. That is, a portion of the lead frame **45** facing the strain gauge portion **31b** of the sensor chip **31** and a peripheral portion thereof is opened as the aperture **45a**.

[0043] A portion of the sensor chip **31** released by the aperture **45a** (that is, a portion facing the aperture **45a**) is a diaphragm portion **31a** which is a deformation portion. The lead frame **45** includes the aperture **45a**, so that the strain generated by the compression of the strain transmission material **36** is transmitted from the strain transmission material **36** to the diaphragm portion **31a** of the sensor chip **31**, and the diaphragm portion **31a** is bent and deformed. The sensor chip **31** measures the pressure of the system water **21** flowing through the flow channel **33** inside the housing **32** by measuring the strain with the strain gauge portion **31b**.

[0044] The lead frame **45** is provided to extract an output from the sensor chip **31**. A lead portion of the lead frame **45** is electrically connected to an electrode portion of the sensor chip **31** by a bonding wire **46**. A part of the lead frame **45** protrudes to an outside of the housing **32**, and the lead portion extends to a protruding portion. A sealing member **37** seals between the lead frame **45** and the housing **32**. Any member can be used as the sealing member **37**, for example, the strain transmission material **36** may be used as the sealing member **37**, an O-ring may be used, or a gasket or the like formed on the lead frame **45** may be used.

[0045] The sensor chip **31** can be electrically connected to the outside of the housing **32** by the lead portion of the lead frame **45**. In the pressure sensor in the related art, when the sensor chip **31** is provided inside the housing **32**, it is difficult to electrically connect the sensor chip **31** to the outside of the housing **32** by connecting a wire to the sensor chip **31**. In the pressure sensor **15** according to the present embodiment, the sensor chip **31** can be electrically connected to the outside of the housing **32** by the lead frame **45** electrically connected to the sensor chip **31**.

[0046] The characteristics of the pressure sensor **15** according to the present embodiment are that the strain transmission material **36** includes the pressure-receiving surface **36a** that is in contact with the fluid flow channel **33** and receives the pressure from the fluid, and the bonding surface **36b** that is positioned at the opposite side of the strain transmission material **36** from the pressure-receiving surface **36a** and is in contact with the inner surface of the housing **32** (that is, the wall surface of the flow channel **33**), and that the entire sensor chip **31** is embedded in the strain transmission material **36**.

[0047] Due to the characteristic that the bonding surface **36b** is positioned at the opposite side of the strain transmission material **36** from the pressure-receiving surface **36a**, in the strain transmission material **36**, when the pressure-receiving surface **36a** receives the pressure from the fluid, a compressive stress is generated in the bonding surface **36b**. That is, in the strain transmission material **36**, when the pressure-receiving surface **36a** receives the pressure from the fluid, a force is generated in a direction in which the bonding surface **36b** is pressed against the

housing **32**, and thus in principle, the bonding surface **36b** does not peel off from the housing **32**.
[0048] Due to the characteristic that the entire sensor chip **31** is embedded in the strain transmission material **36**, a boundary surface between the sensor chip **31** and the strain transmission material **36** is not exposed to the system water **21** or the outside air. Therefore, the decrease in the bonding force between the sensor chip **31** and the strain transmission material **36** due to the humidity of the system water **21** or the outside air does not occur.

[0049] Accordingly, even when receiving the pressure from the fluid, the pressure sensor **15** according to the present embodiment can prevent a decrease in sensitivity of the sensor chip **31** due to the peeling off and a leakage of the fluid from a portion to which the sensor chip **31** is bonded.

[0050] FIG. **5** is an exploded perspective view showing only the sensor chip **31**, the bonding wire **46**, and the lead frame **45** in the pressure sensor **15** according to the present embodiment. An electrode portion **31c** of the sensor chip **31** is electrically connected to a lead portion **45b** of the lead frame **45** by the bonding wire **46**.

[0051] The aperture **45a** provided in the lead frame **45** preferably has a shape having a longitudinal direction and a lateral direction, for example, a rectangular shape, an elliptical shape, or a rounded rectangular shape. FIG. **5** shows, as an example, the aperture **45a** having the rectangular shape in which the longitudinal direction is the flow channel direction and the lateral direction is the horizontal direction.

[0052] The diaphragm portion **31a** of the sensor chip **31** is a portion of the sensor chip **31** that is released by the aperture **45a**. For this reason, a shape of the diaphragm portion **31a** is determined according to the shape of the aperture **45a**, and is preferably a shape having the longitudinal direction and the lateral direction, for example, a rectangular shape, an elliptical shape, or a rounded rectangular shape.

[0053] When the diaphragm portion **31a** having the longitudinal direction and the lateral direction is deformed by receiving the pressure from the system water **21**, a difference in a magnitude occurs between strain in the longitudinal direction and strain in the lateral direction in the strain gauge portion **31b**.

[0054] Where the diaphragm portion **31a** has the shape having the longitudinal direction and the lateral direction, it is preferable to use, as the sensor chip **31**, a sensor chip that detects the respective strain in the longitudinal direction and the lateral direction of the strain gauge portion **31b** and outputs a difference in the strain therebetween. By using such a sensor chip **31** capable of detecting anisotropy of the strain, the pressure sensor **15** can improve pressure detection sensitivity. An existing sensor chip can be used as such a sensor chip **31**.

[0055] In addition, when such a sensor chip **31** is used, it is possible to reduce an influence of thermal strain that generates isotropic strain, that is, an influence of a temperature change. Accordingly, the pressure sensor **15** can reduce a change in output characteristics due to the temperature change.

[0056] It is preferable that the strain transmission material **36** is made of a material having rigidity sufficiently smaller than rigidity of the housing **32**. In general, the housing **32** is made of a material having sufficiently high rigidity, so that deformation due to the pressure of the system water **21** is sufficiently small. The strain transmission material **36** has rigidity sufficiently smaller than that of the housing **32**, so that only the strain transmission material **36** is efficiently deformed due to a water pressure of the system water **21**. Accordingly, the pressure sensor **15** can improve the pressure detection sensitivity.

Embodiment 2

[0057] A pressure sensor according to Embodiment 2 of the invention will be described with reference to FIG. **6**. The pressure sensor **15** according to the present embodiment has the same configuration as the pressure sensor **15** according to Embodiment 1, but is different from the pressure sensor **15** according to Embodiment 1 in that the sensor chip **31** includes a cap member. Hereinafter, the pressure sensor **15** according to the present embodiment will be described mainly

with respect to the differences from the pressure sensor **15** according to Embodiment 1.

[0058] FIG. **6** is an exploded perspective view showing only the sensor chip **31**, the bonding wire **46**, the lead frame **45**, and a cap member **34** in the pressure sensor **15** according to the present embodiment.

[0059] The cap member **34** is a member that is provided on the surface of the sensor chip **31** and covers and protects the strain gauge portion **31b**. The cap member **34** is provided on the surface of the sensor chip **31** opposite a surface in contact with the lead frame **45** in the upper-lower direction, and is embedded in the strain transmission material **36**. The cap member **34** may be made of, for example, silicon, and is bonded to the sensor chip **31** using an adhesive member such as a thermosetting adhesive.

[0060] The cap member **34** includes a cavity **34a**. The cavity **34a** is a recess provided in a surface of the cap member **34** in contact with the sensor chip **31** and surrounding a periphery of the strain gauge portion **31b**. A portion of the sensor chip **31** facing the cavity **34a** is deformable, and the deformable portion functions as the diaphragm portion **31a**. The cavity **34a** preferably has a shape having the longitudinal direction and the lateral direction, for example, a rectangular shape, an elliptical shape, or a rounded rectangular shape. FIG. **6** shows, as an example, the cavity **34a** having the elliptical shape (oval) in which the longitudinal direction is the flow channel direction and the lateral direction is the horizontal direction.

[0061] An outline of the cavity **34a** when the cavity **34a** is projected onto the sensor chip **31** is referred to as an outer shape projection line **51** of the cavity **34a**. An outline of the aperture **45a** when the aperture **45a** is projected onto the sensor chip **31** is referred to as an outer shape projection line **50** of the aperture **45a**. An outline of the cap member **34** when the cap member **34** is projected onto the sensor chip **31** is referred to as an outer shape projection line **52** of the cap member **34**. In the pressure sensor **15** according to the present embodiment, the outer shape projection line **51** of the cavity **34a** is positioned closer to a strain gauge portion **31b** side (inside) than the outer shape projection line **50** of the aperture **45a**, and the outer shape projection line **52** of the cap member **34** is positioned more inward than the outline **53** of the sensor chip **31**.

[0062] The sensor chip **31** included in the pressure sensor **15** according to the present embodiment is small, and is generally manufactured by using a semiconductor process to dice a sensor chip formed in a lattice shape on a silicon wafer into a rectangular shape. Since the outer shape projection line **52** of the cap member **34** is positioned more inward than the outline **53** of the sensor chip **31**, it is possible to adopt a manufacturing method in which the cap member **34** is formed by a semiconductor process on the wafer on which the sensor chip is formed and then is diced.

Accordingly, an external dimension and an arrangement position of the cap member **34** can be determined with high accuracy using the semiconductor process.

[0063] The outer shape projection line **51** of the cavity **34a** is positioned more inward than the outer shape projection line **50** of the aperture **45a**, so that the shape of the diaphragm portion **31a** is determined by the outer shape projection line **51** of the cavity **34a**. The cap member **34** can be manufactured with higher accuracy by the semiconductor process. Therefore, in the pressure sensor **15** according to the present embodiment, a manufacturing variation of the shape of the diaphragm portion **31a** determined by the shape of the cavity **34a** of the cap member **34** can be extremely reduced. Accordingly, the variation in the sensitivity of the sensor chip **31** due to the variation in the shape of the diaphragm portion **31a** can also be reduced.

[0064] The pressure sensor **15** according to the present embodiment has an effect of reducing variations in the sensitivity of the sensor chip **31** in addition to the effect of the pressure sensor **15** according to Embodiment 1.

[0065] The invention is not limited to the above-described embodiments and may include various modifications. For example, the embodiments have been described in detail to facilitate understanding of the invention, and the invention is not necessarily limited to those including all configurations described above. A part of configurations of one embodiment can be replaced with a

configuration of another embodiment. A configuration of another embodiment can be added to a configuration of one embodiment. A part of a configuration in each embodiment may be deleted, or may be added with or replaced with another configuration.

REFERENCE SIGNS LIST

[0066] **1**: dispensing device [0067] **2**: nozzle [0068] **4**: syringe pump [0069] **5**: solenoid valve [0070] **6**: gear pump [0071] **7**: tank [0072] **8**: pipe [0073] **9**: container [0074] **10**: plunger [0075] **11**: ball screw [0076] **12**: drive motor [0077] **13**: sample dispensing mechanism [0078] **14**: control unit [0079] **15**: pressure sensor [0080] **15a**: pressure sensor of related art [0081] **16**: arm [0082] **21**: system water [0083] **22**: liquid [0084] **23**: dividing air [0085] **31**: sensor chip [0086] **31a**: diaphragm portion [0087] **31b**: strain gauge portion [0088] **31c**: electrode portion [0089] **32**: housing [0090] **32a**: upper housing [0091] **32b**: lower housing [0092] **32c**: contact surface between upper housing and lower housing [0093] **33**: flow channel [0094] **34**: cap member [0095] **34a**: cavity [0096] **35**: flow channel [0097] **36**: strain transmission material [0098] **36a**: pressure-receiving surface [0099] **36b**: bonding surface [0100] **37**: sealing member [0101] **40**: bonding agent [0102] **45**: lead frame [0103] **45a**: aperture [0104] **45b**: lead portion [0105] **46**: bonding wire [0106] **50**: outer shape projection line of aperture [0107] **51**: outer shape projection line of cavity [0108] **52**: outer shape projection line of cap member [0109] **53**: outline of sensor chip

Claims

1. A pressure sensor comprising: a housing having therein a space serving as a fluid flow channel; a sensor chip including a strain detection unit and formed of a thin-film semiconductor; and a strain transmission material provided inside the housing, wherein the strain transmission material includes a pressure-receiving surface in contact with the space, and a bonding surface positioned at an opposite side of the strain transmission material from the pressure-receiving surface and fixed to an inner surface of the housing, and the entire sensor chip is embedded inside the strain transmission material.
 2. The pressure sensor according to claim 1, wherein the strain transmission material has rigidity smaller than that of the housing.
 3. The pressure sensor according to claim 1, further comprising: a lead frame provided in contact with the sensor chip and electrically connected to an electrode of the sensor chip, wherein the lead frame includes an opening that opens in a range facing a portion of the sensor chip including the strain detection unit.
 4. The pressure sensor according to claim 3, further comprising: a cap member provided on a surface of the sensor chip opposite a surface in contact with the lead frame, wherein the cap member includes a recess having a shape having a longitudinal direction and a lateral direction and surrounding a periphery of the strain detection unit on a surface in contact with the sensor chip, and when the recess, the opening, and the cap member are projected onto the sensor chip, an outline of the projected recess is positioned more inward than an outline of the projected opening, and an outline of the projected cap member is positioned more inward than an outline of the projected sensor chip.
 5. The pressure sensor according to claim 3, wherein a part of the lead frame is embedded in the strain transmission material, and another part of the lead frame protrudes to an outside of the housing.
-